Example 1: United Nations Population Data Exercise

United Nations | Data Portal Population Division

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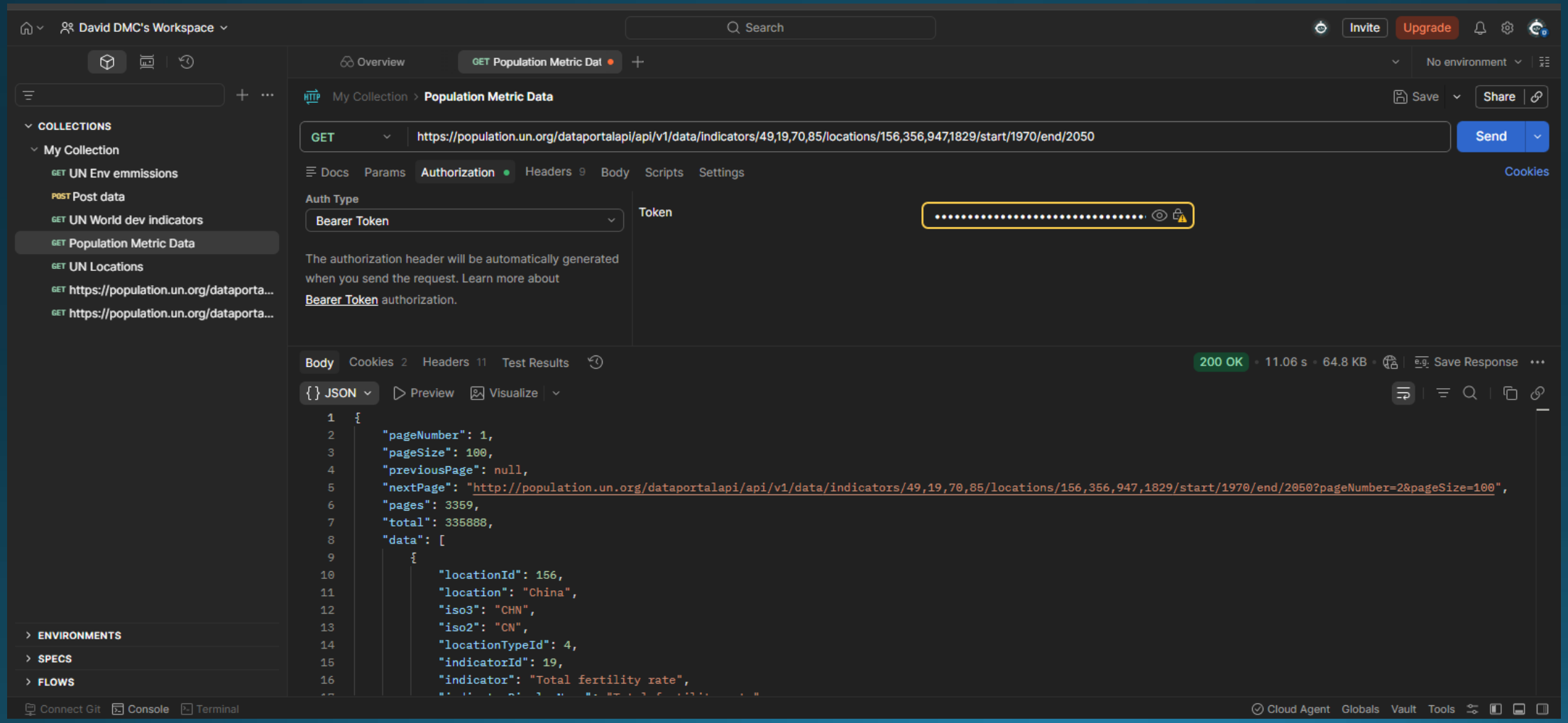
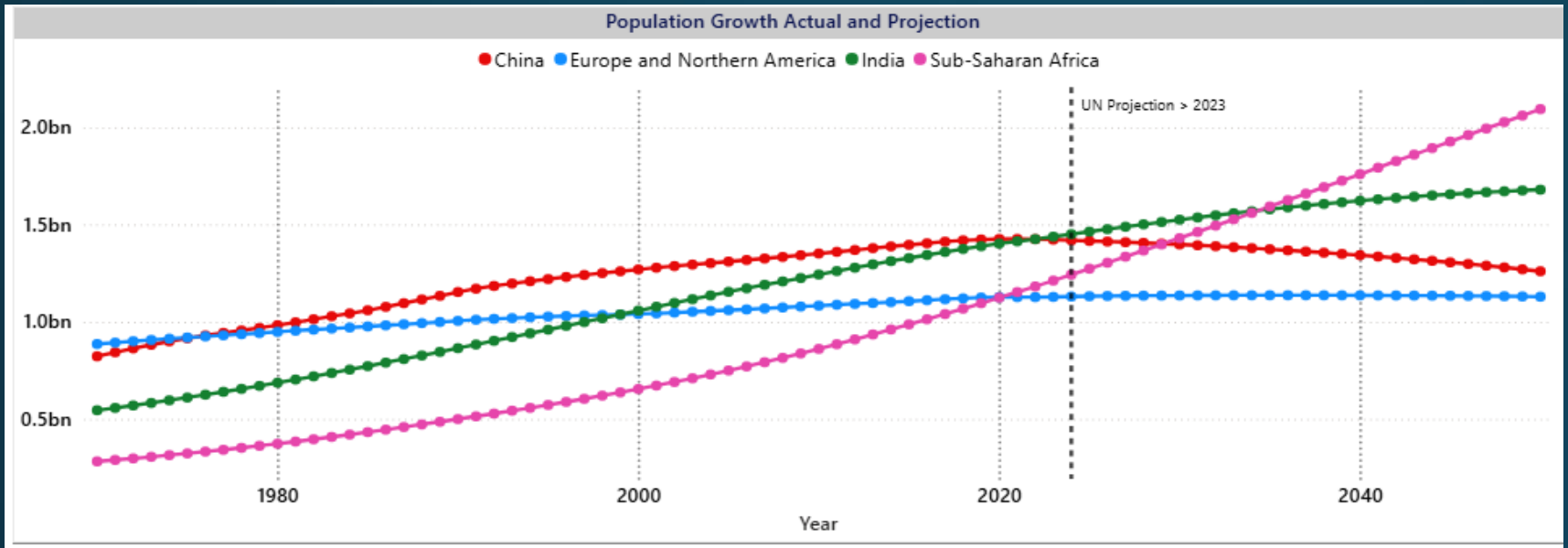
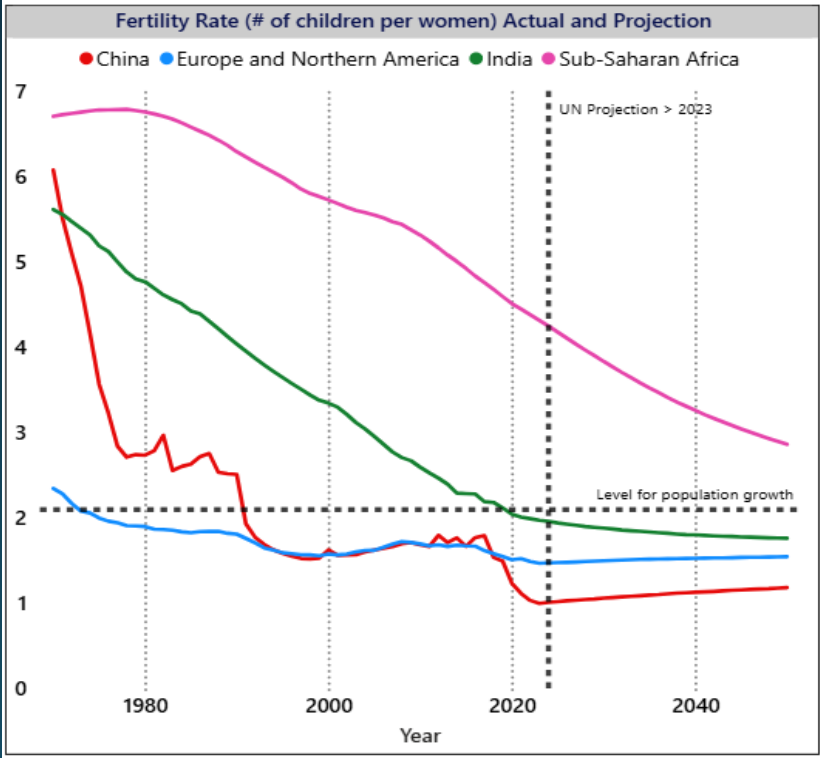
The UN Population data portal is a rich resource for all sorts of Population data. Data is available through a REST API, with clear API documentation. I decided to test a number of the APIs using Postman (image below) and then pulled the data into Power BI.

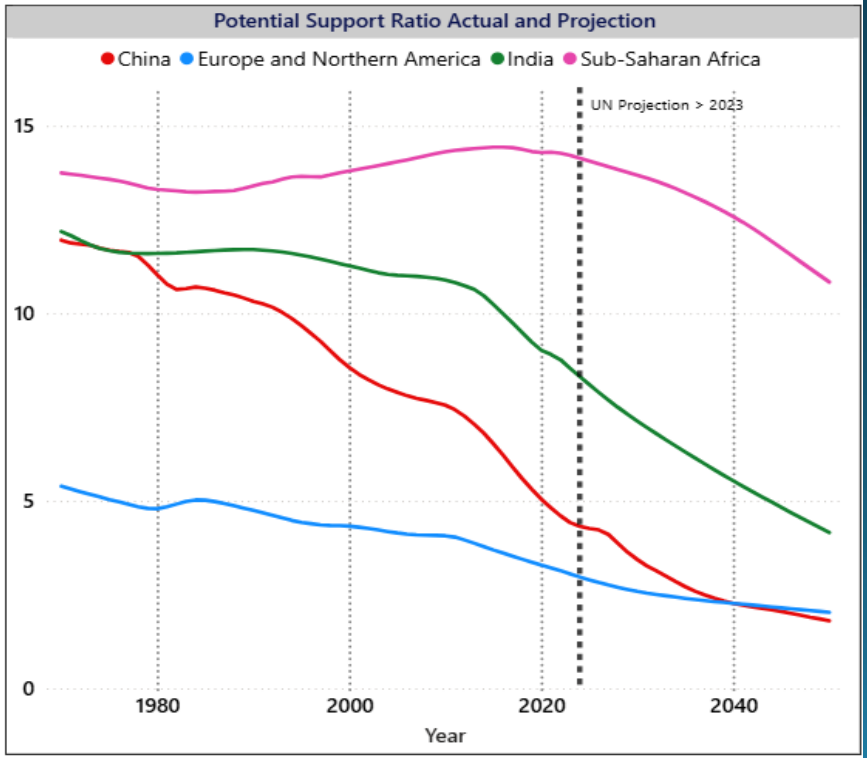


Source: *United Nations Population Division, New York, World Population Prospects: The 2023 edition*



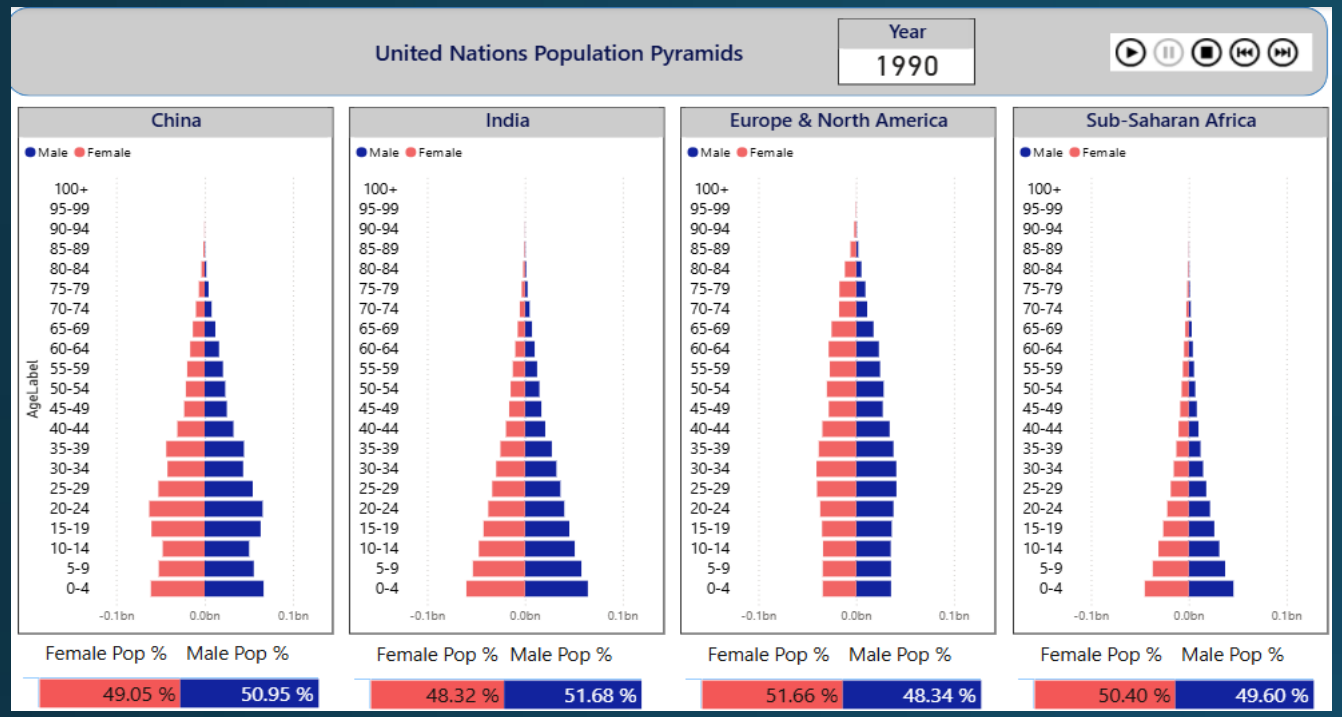
** Actual Data is available to 2023. Beyond 2023, the data is based on a projection*

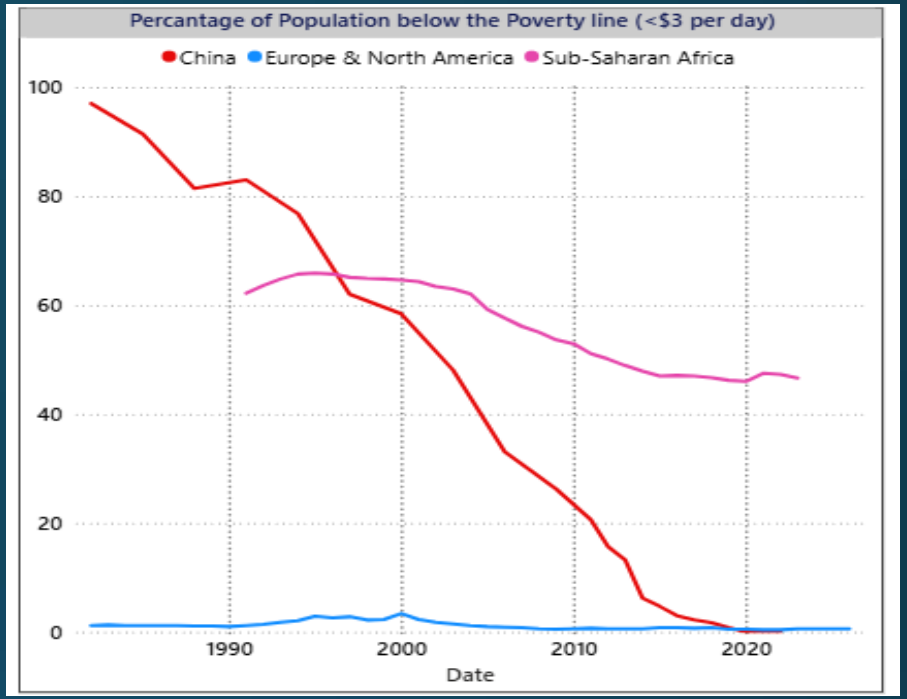




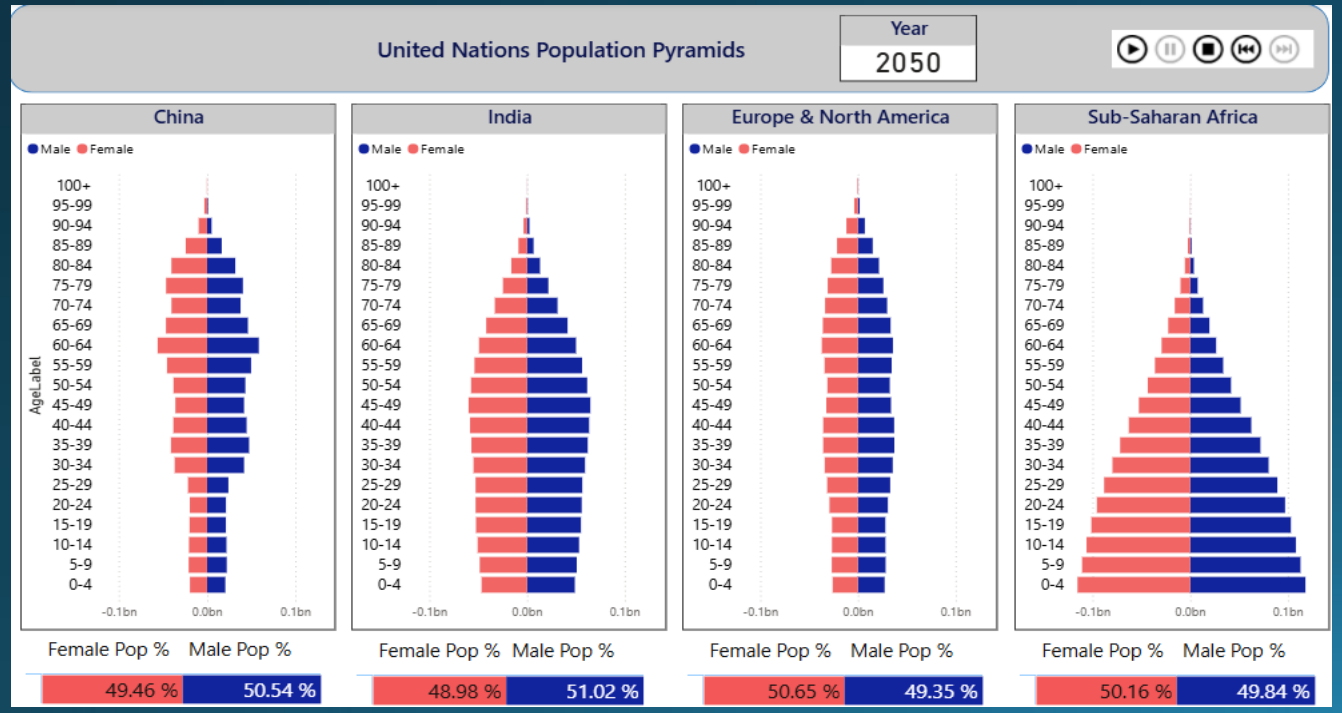
** The Fertility rate shows the average number of births per women. The replacement rate is 2.1, which is the required level required for populations to grow.*

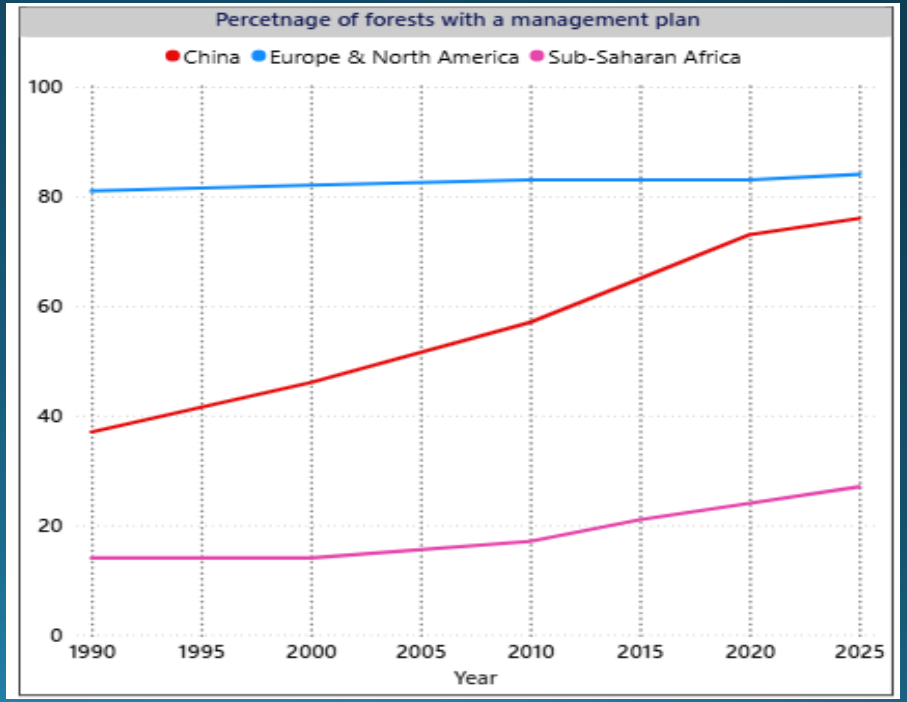
** The potential support ratio (PSR) is a demographic metric measuring the number of working-age people (ages 20–64) available per older person (aged 65+). A higher ratio indicates a larger workforce to support retirees, while a falling ratio signals increased strain on pension and healthcare systems*





*This indicator measures the Proportion of the population who exist on less than $3 USD per day (%)





*This indicator measures progress towards Sustainable Forest Management (SFM) as a **Proportion of forest area with a long-term management plan (%)***

Key Takeaways and Insights

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1. The global population is still growing, albeit at a reduced rate. China, which was once the world's most populous country, now sits second in global population counts behind India, as decades of China's 'One Child Policy' has led to a declining and aging population.

2. The average number of live births per women over a lifetime has fallen markedly in many regions over the past several decades. Most regions have a fertility rate below 2.1, which is the required level required for populations to grow. The exception is Sub-Saharan Africa.

3. The population pyramids can visually demonstrate the changes in demographic composition between 1990 and the 2050 projection. China's aging population can be observed by an almost inverted pyramid shape by 2050 as the number of people below 30 are dwarfed by those older, with males outnumbering females by 51% - 49%. For economic planners, the Potential Support Ration is a declining and concerning metric as the number of people of working age (20-64) vs those 65+ is projected to continuously fall as we head towards 2050.

4. The population growth will be concentrated in Sub Saharan Africa in the coming decades and is expected to surpass 2 billion by mid 2040's. It will become the world's most populous region, overtaking South - East Asia.

Population Growth in Sub-Sharan Africa poses a real challenge, as the region is exposed to climatic extremes, a warming climate and associated crop sensitivity. Ensuring adequate nutrition for a growing population will be a huge challenge, especially when close to 50% of the population in 2023 live below the poverty line (Income of $3 per day). How are governments going to prepare for this? What will be the impact, not just on unemployment, but environmentally? Only 30% of the region's forests are protected by some form of sustainable management plan, so is there a risk of increased deforestation in the region as land pressure builds?